

AssetDNA: Enterprise AI Industrial Platform

Comprehensive Project Report

1. Executive Summary

AssetDNA is an AI-first industrial knowledge intelligence platform designed to serve as the "Living Memory of Every Industrial Asset." In modern industrial operations, telemetry data, maintenance logs, and engineering manuals are highly fragmented. AssetDNA unifies these data streams, leveraging advanced Large Language Models (LLMs) to synthesize context and deliver instant, deterministic root-cause analysis and actionable insights.

2. The Problem

Heavy industries (manufacturing, energy, utilities) suffer from severe data silos:

- **SCADA/IoT Systems:** Produce massive volumes of raw telemetry (vibration, temperature) without operational context.
- **CMMS (Computerized Maintenance Management Systems):** Hold historical work orders, but are notoriously difficult to search.
- **Unstructured Documentation:** Millions of pages of OEM manuals, schematics, and standard operating procedures (SOPs) are stored as static PDFs. When an alarm triggers, engineers must manually cross-reference these distinct systems to diagnose the failure, leading to prolonged unplanned downtime.

3. The Solution

AssetDNA solves this by establishing a unified knowledge graph for every physical asset.

- **Live Telemetry:** Connects to asset sensors for real-time status.
- **Document Intelligence:** Vector-indexes OEM manuals and engineering schematics.
- **Maintenance History:** Aggregates decades of work orders and inspections.
- **AI Synthesizer:** When an engineer asks, "Why did Pump PX-100 fail?", AssetDNA processes the telemetry against the manuals and historical logs to instantly answer with the root cause, backed by clickable, deterministic citations.

4. Key Features

4.1. Mission Control Dashboard

A high-level view of the entire fleet. Displays live KPIs including Total Assets, Uptime, Critical Assets, and Maintenance Due. The dashboard immediately surfaces AI Briefings—highlighting critical anomalies and high-risk assets before they fail.

4.2. Asset Workspace & Explorer

Every machine has a dedicated digital twin workspace containing:

- Live Status & Health Summaries
- Consolidated Maintenance Logs
- Relevant Engineering Changes
- Instant Document Retrieval

4.3. AI Investigation Engine

A sophisticated chat interface built directly into the asset workspace. Engineers query the asset in natural language. The AI generates deep technical responses, explicitly citing the source documents (e.g., "OEM Manual Section 4.2") to eliminate hallucination.

4.4. Fleet-Wide Knowledge Assistant

A global AI agent capable of answering cross-system queries (e.g., "Which cooling towers require lubrication this month?") by aggregating data across the entire asset registry.

4.5. Proactive Maintenance Planner

Translates AI insights into actionable maintenance schedules, bridging the gap between diagnosis and physical resolution.

5. Technical Architecture

The platform is built on a modern, decoupled, serverless architecture optimized for high performance and AI scalability.

5.1. Frontend Architecture (Next.js)

- **Framework:** React 18, Next.js (App Router)
- **Styling:** TailwindCSS, Vanilla CSS (Glassmorphism, Dark-mode SaaS aesthetics)
- **State Management:** React Context API, Custom Hooks (`useApi`, `useWorkspace`)
- **Animation:** Custom Framer-like animation primitives (`AnimatedSlideUp`, `AnimatedScale`)
- **Data Fetching:** Axios with robust retry policies and intelligent mock data fallback injection.

5.2. Backend Architecture (Express / Node.js)

- **Framework:** Express.js running on Node 18
- **Database:** Google Cloud Firestore (NoSQL, document-based architecture)
- **Design Pattern:** Repository-Service-Controller pattern for strict separation of business logic and data access.
- **AI Integration:** Groq SDK for ultra-low latency LLM inference.
- **Security:** Firebase Authentication middleware.

6. Deployment & Future Scalability

Currently operating with a robust local demo engine that seamlessly intercepts API failures to guarantee a rich UI experience during presentations. The architecture is inherently cloud-native, designed to scale globally via Google Cloud Functions and Firebase, ensuring millisecond latency for enterprise telemetry ingestion.

7. Conclusion

AssetDNA represents a paradigm shift in industrial reliability. By transforming static maintenance data into a living intelligence engine, it empowers engineering teams to predict, diagnose, and resolve industrial issues faster than ever—ultimately saving millions in unplanned downtime.